

ASSESSMENT SCHEDULE 2022
Mathematics and Statistics 91261 (2.6): Apply algebraic methods in solving problems

One	Expected Coverage	Achievement (u)	Merit (r)	Excellence (t)
(a)	$\frac{(x^2)^2}{(2x^3y^2)^2}$ $= \frac{1}{4x^2y^4}$	Correct expression, no decimals allowed		n1: correctly dealing with negative indice eg line 1
(b)	$= 15x^2 - 10x - 12x + 8$ $= -5x(-3x + 2) + 4(-3x + 2)$ $= (2 - 3x)(4 - 5x)$ $f = 4 - 5x$	$f = (4 - 5x)$ clearly identified		n1: correctly factorising but not concluding ie line 3, could also be factorised $(5x-4)(3x-2)$
(c)	$\log \sqrt[3]{10}$ $= \log(10)^{\frac{1}{3}}$ $= \frac{1}{3} \log(2 \times 5)$ $= \frac{1}{3}(\log 2 + \log 5)$ $= \frac{1}{3}(x + y)$	Cube root converted to indice AND 10 split into 2 and 5	Correct answer Any equivalent solution accepted	n1: Cube root converted to indice OR 10 split into 2 and 5
(d)	$p = \sqrt{\frac{2-ax^2}{x^2-b}}$ $p^2 = \frac{2-ax^2}{x^2-b}$ $p^2(x^2-b) = 2-ax^2$ $p^2x^2 - p^2b = 2-ax^2$ $p^2x^2 + ax^2 = 2 + p^2b$ $x^2 = \frac{2 + p^2b}{p^2 + a}$ $x = \pm \sqrt{\frac{2 + p^2b}{p^2 + a}}$	Equation partially rearranged to line 4 $p^2x^2 - p^2b = 2 - ax^2$	Equation rearranged correctly with final answer including +/-	
(e)	$b^2 - 4ac > 0$ $(m+2)^2 - 4(4m-7) > 0$ $m^2 - 12m + 32 > 0$ $(m-4)(m-8) > 0$ $\therefore m < 4, m > 8$	Correct substitution into $b^2 - 4ac > 0$ Or equivalent (accept $b^2 - 4ac = 0$)	Quadratic $m^2 - 12m + 32 > 0$ stated (or equivalent)	T1: Correct and clear working and correct final solution given
(f)	$\text{LHS} =$ $\left(x^{\frac{1}{2}} + x^{-\frac{1}{2}}\right)^2 \left(x^{\frac{1}{2}} - x^{-\frac{1}{2}}\right)^2 + 4$ $= (x + 2 + x^{-1})(x - 2 + x^{-1}) + 4$ $= x^2 + 2 + \frac{1}{x^2} = \text{RHS}$	Expansion of $\left(x^{\frac{1}{2}} + x^{-\frac{1}{2}}\right)^2$ OR $\left(x^{\frac{1}{2}} - x^{-\frac{1}{2}}\right)^2$	Expansion and simplification of $\left(x^{\frac{1}{2}} + x^{-\frac{1}{2}}\right)^2$ AND $\left(x^{\frac{1}{2}} - x^{-\frac{1}{2}}\right)^2$	T1: Showing final expansion of LHS including + 4 T2: LHS fully simplified to match RHS (or equivalent) and statement made to justify LHS = RHS

N0	N1	N2	A3	A4	M5	M6	E7	E8
No response; no relevant evidence	ONE partial solution.	1 of u	2 of u	3 of u	1 of r	2 of r	T1	T2 or two T1

Two	Expected Coverage	Achievement (u)	Merit (r)	Excellence (t)
(a)	$= 3\log x - \log x - 5\log x$ $= -3\log x$	States $-3\log x$ or any equivalent		n1: on log rule used correctly
(b)	$f(x) = (x - 5)^2 + 4 - 25$ $f(x) = (x - 5)^2 - 21$	Correct answer stated		n1: a or b correct
(c) (i)	$V = 52 \times 0.8^t$ $24 = 52 \times 0.8^t$ $\log\left(\frac{24}{52}\right) = t \times \log 0.8$ $t = \frac{\log\left(\frac{24}{52}\right)}{\log 0.8} \approx 3.465$ After 3.5 years (or during 2017) also accept after 4 years	Correct working to line 3	Correct answer in years with algebraic working <i>Do not accept after 3 years</i> Accept 3.465 years	
(c) (ii)	$52 \times 0.8^t = 39.5 \times 0.85^t$ $\frac{52}{39.5} = \frac{0.85^t}{0.8^t} = \left(\frac{0.85}{0.8}\right)^t$ $t = \frac{\log\left(\frac{52}{39.5}\right)}{\log\left(\frac{0.85}{0.8}\right)} \approx 4.535$ In the year 2018.	Equation set up	Rearranged into index equation: $a = b^t$ as in Line 2. If equation was set up with 39 instead of 39.5 but followed through correctly after that award r	T1: Solved algebraically and year stated (2018).
(d)	$0 = 0.0024 + 0.018 - c + d$ $c - d = 0.0204$ (1) $0.43 = 0.0024(-5)^3 + 0.018(-5)^2 + 5c + d$ $5c + d = 0.28$ (2) Solve 1 & 2 simultaneously $c = 0.05, d = 0.0296$ $f(x) = 0.0024x^3 + 0.018x^2 + 0.05x - 0.0296$	Forms either eq (1) or eq (2)	Forms eq (1) and eq (2)	T1: Finds the correct values for c and d using algebraic methods not calculus T2: States the final correct function

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Three	Expected Coverage	Achievement (u)	Merit (r)	Excellence (t)
(a)	$\frac{x(x+3)+2(x+5)}{(x+5)(x-3)}$ $= \frac{x^2-x+10}{(x+5)(x-3)}$	Correct simplified expression <i>denominator can be expanded</i>		n1: correctly combined into single fraction, or two fractions with common denominator
(b)	$5x^2 - 72 = 2x(x - 3)$ $3(x^2 + 2x - 24) = 0$ $3(x + 6)(x - 4) = 0$ $x = -6 \text{ or } x = 4$	Finds 2 roots		
(c)	$7a^2 + 5a + 2 = 0$ $\Delta = 25 - 4 \times 7 \times 2$ $\Delta = -31$ $\Delta < 0$ therefore no real roots	Finds the discriminant	States that there are no real roots with justification	
(d)	$x^2 - 3x - 2 = 6$ $x^2 - 3x - 8 = 0$ $x = \frac{3 \pm \sqrt{41}}{2}$	Quadratic solved but decimal solutions given $x=4.70156$ or $x=-1.70156$ OR CAO	Quadratic formula or completing the square used and answer kept in surd form	n1: correct rearrangement to line 2
(e)	$3(x - 6)(x - 13) = 1620$ $(x - 6)(x - 13) = 540$ $x^2 - 19x - 462 = 0$ $x = 33, -44$ Length is 33cm and width = 26cm	Forms the correct initial equation	Solves to find $x = 33, -44$	T1: Eliminates negative solution and states the correct length and width (including units)
(f)	$\Delta = (5k + 7)^2 - 4(3k + 6)(3k + 1) > 0$ $-11k^2 - 14k + 25 > 0$ $11k^2 + 14k - 25 < 0$ $(k - 1)(11k + 25) < 0$ $\therefore \frac{-25}{11} < k < 1$	$-11k^2 - 14k + 25 > 0$ Or equivalent eg $-11k^2 - 14k + 25 = 0$	$(k - 1)(11k + 25) < 0$ stated with correct inequality sign or equivalent	T1: Reference to $\frac{-25}{11} < k < 1$ made T2: The justification for $\frac{-25}{11} < k < 1$ is made clear through working or a diagram

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Cut Scores (based on 2021 Exam)

Not Achieved	Achievement	Achievement with Merit	Achievement with Excellence
0 – 7	8 – 13	14 – 19	20 – 24